

Astro HW 11

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11.1

a)

11.2 Spiral galaxies - Tracing the Matter

a)

Given...

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$
$$v_{esc} = 500 \text{ km s}^{-1} \quad 8.5 \text{ kpc}$$

We find that ...

$$v_{esc} = \frac{\sqrt{2GM}}{\sqrt{R}}$$
$$\frac{v_{esc}\sqrt{R}}{\sqrt{2G}} = \sqrt{M}$$
$$M = 2.470 \times 10^{11} \text{ M}_{\odot}$$

b)

i)

Given ...

$$M_V = -2.76 \log_{10} \left(\frac{P}{10 \text{ d}} \right) - 4.16$$
$$m_V = 18.9, \quad P = 31.47 \text{ d}$$
$$m - M = 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right) \rightarrow d = 10 \exp \left(\frac{m - M + 5}{5} \right) \text{ pc}$$

We find ...

$M_v = -5.534, \quad d = 7.706 \times 10^{-1} \text{ Mpc}$
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ii)

53.9 km/s

iii)

$$g = 3.914 \times 10^{-13} \text{ km/s}$$

iv)

Given ...

$$g = GM/R^2$$

$$g_{m31} = GM_{m31}/R^2, \quad g_{MW} = GM_{MW}/R^2$$

$$R = 7.706 \times 10^{-1} \text{ Mpc}$$

We can find ...

$$g_{m31} = 2g_{MW}$$

$$3.914 \times 10^{-13} = 2GM_{MW}/R^2$$

$$\frac{(3.914 \times 10^{-13})R^2}{2G} = M_{MW}$$

$$M_{MW} = 8.339 \times 10^{11} \text{ M}_{\odot}$$

11.3